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Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

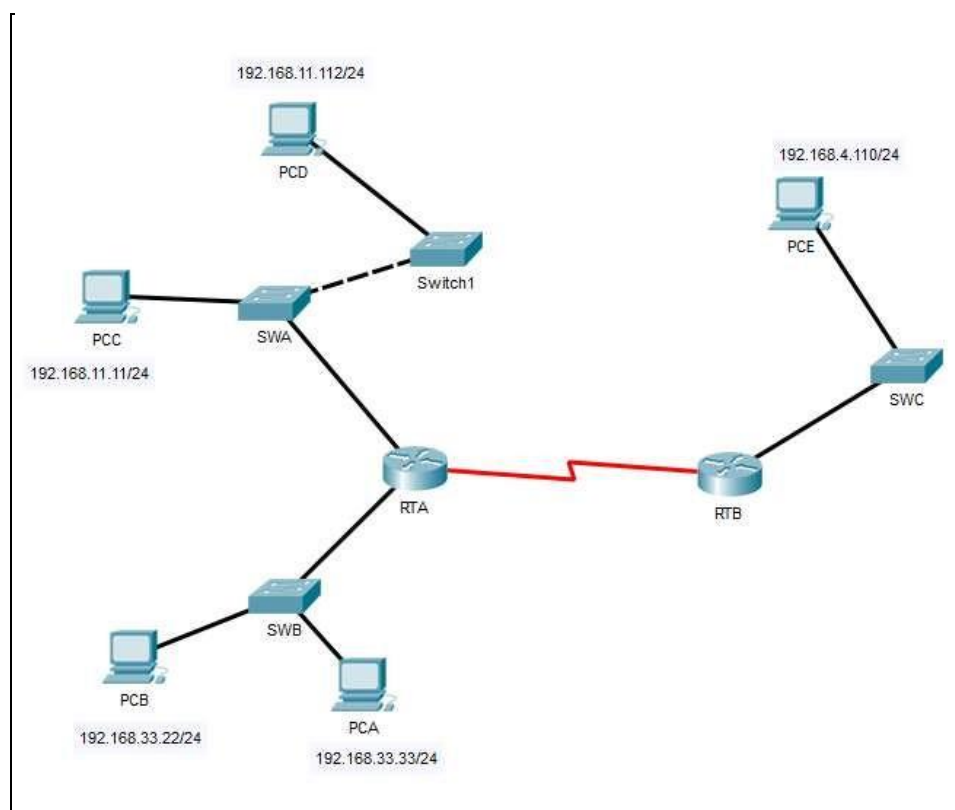


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1      -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1      -         000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1      -         0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
      1    0002.4a00.0e91    DYNAMIC     Fa0/1
      1    000c.8546.7d85    DYNAMIC     Fa1/1
SWA#
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    000c.cf0c.593a    DYNAMIC Fa0/1
SWB#|
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
SWC#|
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCB

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
C:\>arp -a
No ARP Entries Found
C:\>
```

- e. What are your thoughts on the results?

There is no MAC addresses on each PC yet.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click PCA and click the Desktop tab > Command Prompt.
- Enter the ping 192.168.33.22 command. This may take a few seconds.
- In the Command prompt of PCA, type arp -a. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
C:\>
```

- In the Command prompt of PCB, type arp -a. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic
C:\>
```

- In the Command prompt of PCC, PCD abd PCE, type arp -a. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click PCC and click the Desktop tab > Command Prompt.
- b. Enter the ping 192.168.33.22 command (ping to PCB). Then type arp -a. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
C:\>
```

- c. Enter the ping 192.168.11.112 command (ping to PCD). Then type arp -a. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- d. Enter the ping 192.168.4.110 command (ping to PCE). Then type arp -a. Paste the result after these commands here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.

When PCC ping PCD and PCB , the PC now has MAC addresses. But when PCC tried to ping PCE, which from a different router and subnet it fails to resolved MAC address because PCE has a different subnet.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     20         00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     20         0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

- g. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
```

```
RTB#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     20         00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     20         0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable
```

```
SWA#show arp
```

```
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91    DYNAMIC Fa0/1
1       000c.8546.7d85    DYNAMIC Fa1/1
-----
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
```

```
SWB#show arp
```

```
SWB#show mac-address-table
```

```
SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
-----
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
```

```
SWC#show arp
```

```
SWC#show mac-address-table
```

```
SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614    DYNAMIC Fa0/1
```

d. Do switches use arp table? (Y/N)

No

e. Explain your answer in (d) *Hint: the answer may surprise you. Google for the explanation.

It is not part of NetComm syllabi, it is just for knowledge.

Switches does not use arp table because ARP operates at layer 3 but switches operate at level 2

f. What information does the command show mac-address-table gives?

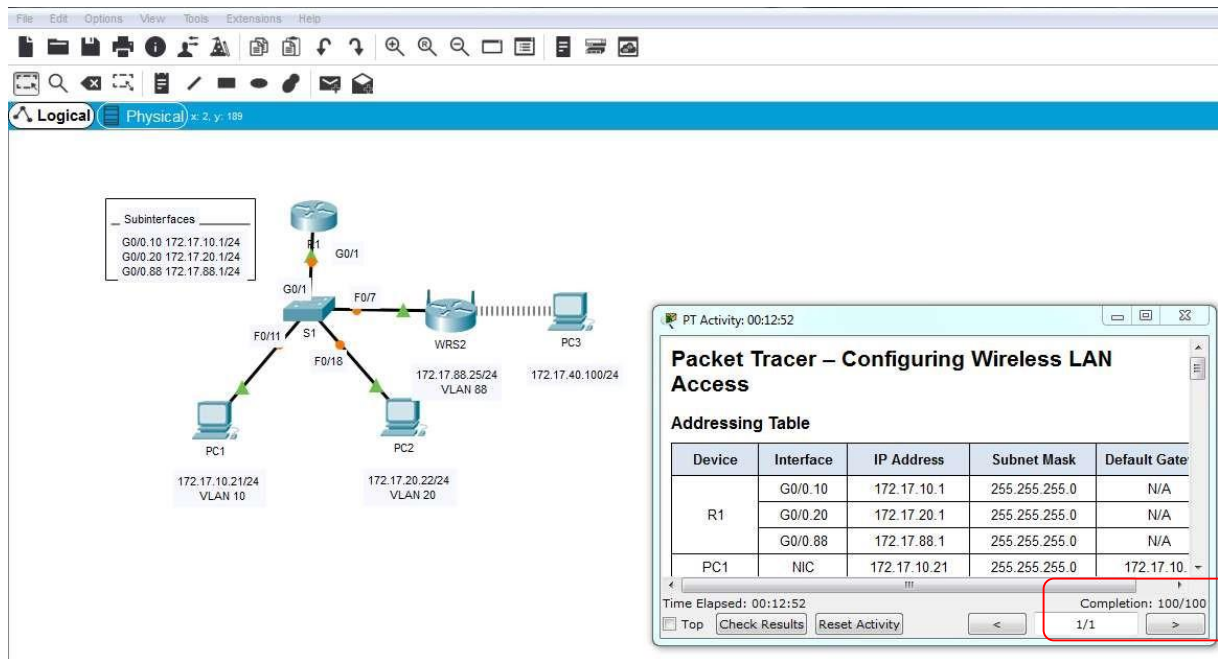
Mac address table, the information given is MAC address, VLAN, port, Type

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- a. Change the Lab 4 filename to include your name. *Example: Lab4AliAhmad.pkt
- b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



- c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - C:/Users/User/Downloads/Lab 4 - Wireless.pka - Lab4AinnurAshikin - 2025-01-20 20:15:43

File Edit Options View Tools Extensions Window Help

Logical Physical x 192, y: 3 Root 02:52:30

Subinterfaces

- G0/0.10 172.17.10.1/24
- G0/0.20 172.17.20.1/24
- G0/0.88 172.17.88.1/24

PC1: 172.17.10.21/24 VLAN 10

PC2: 172.17.20.22/24 VLAN 20

PC3: 172.17.40.100/24

WRS2: 172.17.88.25/24 VLAN 88

PT Activity: 00:26:10

Part 1: Configure a Wireless Router

Step 1: Connect the Internet interface of WRS2 to S1.
Connect the WRS2 Internet interface to the S1 F0/7 interface.

Step 2: Configure the Internet connection type.

- Click WRS2 > GUI tab.
- Set the Internet Connection type to Static IP.
- Configure the IP addressing according to the Addressing Table.

Step 3: Configure the network setup.

- Scroll down to **Network Setup**. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
- Enable the DHCP server.
- Scroll to the bottom of the page and click **Save Settings**.

Time: 00:26:14

Time Elapsed: 00:26:10

Completion: 100/100

Check Results Back 1/1 Next

New delet PDU List \

Conner Straight-Through